For your dataset, I would **not recommend a conventional deep neural network** because of the ratio of samples to outputs:

Property	Value
Number of samples	**39**
Non-invasive features	**10**
Invasive variables (targets)	**1638**

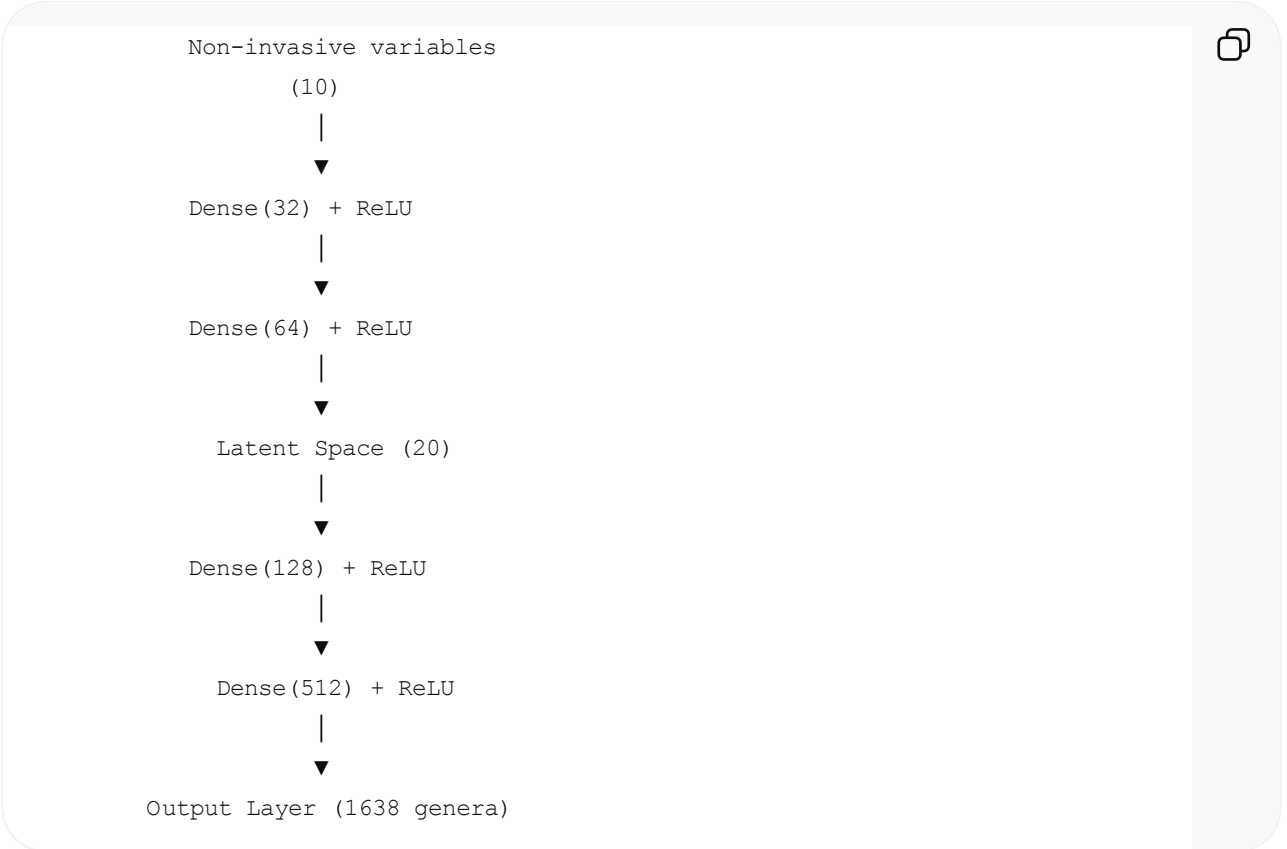
A fully connected neural network with 1638 outputs would have thousands of trainable parameters but only 39 training examples, making severe overfitting almost inevitable.

Recommended Neural Network: Multi-Task Autoencoder Network (MTAN)

Instead of directly learning

$$X_{10} \rightarrow Y_{1638}$$

learn a compressed representation of the invasive variables.



Why this architecture?

1. Multi-output learning

Instead of building 1638 separate models,

```
10 inputs
      |
      ▼
 Shared hidden layers
      |
      ▼
1638 outputs
```

the model learns relationships among microbial genera.

---

2. Bottleneck (latent layer)

The latent layer forces the network to learn

- microbial communities
- co-occurrence patterns
- shared biological structures

instead of memorizing individual genera.

---

3. Regularization

Because the dataset is tiny, use aggressive regularization:

```
Dense(32)
↓

BatchNormalization

↓

Dropout(0.4)

↓

Dense(64)

↓

Dropout(0.4)

↓
```

```
Latent(20)
```

Model Specification

Input

```
10 non-invasive variables
```

Standardize:

```
x = (x - mean) / std
```

Hidden Layer 1

```
Dense(32)
Activation = ReLU
```

Hidden Layer 2

```
Dense(64)
Activation = ReLU
Dropout = 0.4
```

Bottleneck

```
Dense(20)
Activation = Linear
```

This becomes the learned microbiome representation.

Decoder

```
Dense(128)
↓

Dense(512)
↓

Dense(1638)
```

Output activation:

```
Linear
```

(if abundances are log-transformed)

or

```
Softplus
```

(if values are non-negative).

Loss Function

For microbiome abundance data,

instead of MSE,

use

Huber Loss

```
loss = Huber(delta=1)
```

Advantages:

- robust to outliers
- more stable than MSE

Optimizer

```
Adam

learning_rate = 0.0005
```

Regularization

```
L2 weight decay = 0.001

Dropout = 0.4

EarlyStopping(patience=50)
```

Validation

With only 39 individuals:

Leave-One-Out Cross Validation (LOOCV)

```
Train: 38

Test: 1

Repeat 39 times
```

or

Repeated 5-fold CV.

Avoid a simple 80/20 split because the test set would contain only about 8 samples.

Alternative: Autoencoder + Predictor (My Preferred Neural Network)

This is a two-stage approach.

Stage 1

Train an autoencoder only on the invasive data.

```
1638
 |
 ▼
512
 |
 ▼
128
```

```
 |
 ▼
20 latent variables
 |
 ▲
128
 |
 ▲
512
 |
 ▲
1638
```

The encoder learns a compact representation of the microbiome.

Stage 2

Train a small neural network

```
10 non-invasive variables

↓

Dense(32)

↓

Dense(32)

↓

20 latent variables
```

using the latent vectors generated in Stage 1.

Finally,

```
Predicted latent vector

↓

Frozen decoder

↓

Predicted 1638 genera
```

Why this is superior

Instead of learning

```
10  → 1638
```

the predictor only learns

```
10 → 20
```

which is dramatically easier with only 39 samples.

The decoder already knows how latent variables map to the microbiome.

Recommendation for a Research Paper

For this specific dataset (39 subjects, 10 predictors, 1638 invasive measurements), the **Autoencoder + Predictor architecture** is the neural-network approach I would recommend. It combines dimensionality reduction, multi-task learning, and regularization, making it far more appropriate than a conventional multilayer perceptron and more likely to generalize from such a small sample size.